

PHISHING EMAIL DETECTION & AWARENESS REPORT

Future Interns – Cyber Security Internship

Task 2: Phishing Email Detection & Awareness System

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Date: June 2026

Executive Summary

Phishing attacks remain one of the most common cybersecurity threats affecting individuals and organizations. Attackers use fraudulent emails to trick recipients into revealing sensitive information, downloading malicious files, or performing unauthorized actions.

This report analyzes three simulated phishing email samples and identifies indicators commonly used in phishing campaigns. Each email was reviewed for sender authenticity, email header anomalies, social engineering tactics, malicious links, and overall risk level.

The objective of this assessment is to improve awareness and help users recognize and avoid phishing attempts.

Objectives

The purpose of this task is to:

- Analyze phishing email samples
 - Identify phishing indicators
 - Examine email header information
 - Classify email risk levels
 - Explain attack techniques
 - Provide prevention recommendations
 - Improve cybersecurity awareness
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Tools Used

- Microsoft Word
 - Google Message Header Analyzer
 - MXToolbox Email Header Analyzer
 - GitHub
 - Web Browser
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Email Sample Analysis

Sample 1 – Bank Account Alert Phishing Email

Email Scenario

The attacker impersonates a banking institution and claims suspicious activity has been detected on the victim's account.

The email attempts to create fear and urgency by threatening account suspension if immediate action is not taken.

Header Analysis

Field	Value
Sender Domain	bankofamerica-alerts.net
SPF	FAIL
DKIM	NONE
DMARC	FAIL
Mail Server	mail.bankofamerica-alerts.net

Phishing Indicators

Indicator	Observation
Fake Sender Domain	Not an official Bank of America domain
Urgency	Action required immediately
Fear Tactic	Account suspension threat
Credential Harvesting	Requests Online ID and Passcode
Personal Information Request	Requests Social Security Number
Suspicious URL	bankofamerica-secure-verify.net
Authentication Failure	SPF and DMARC failed

Risk Classification

Classification: PHISHING

Risk Level: HIGH

Potential Impact

- Credential theft
 - Identity theft
 - Financial fraud
 - Unauthorized banking access
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Sample 2 – IT Helpdesk Password Reset Phishing Email

Email Scenario

The attacker impersonates an organization's IT department and attempts to trick employees into resetting their passwords using a fraudulent portal.

The email uses urgency and the threat of losing access to company resources.

Header Analysis

Field	Value
Sender Domain	company-support-portal.com
SPF	SOFTFAIL
DKIM	NONE
DMARC	FAIL
Mail Server	smtp.company-support-portal.com

Phishing Indicators

Indicator	Observation
Generic Greeting	Dear Employee
Fake IT Portal	Unofficial support domain
Password Reset Request	Requests immediate action
Threat Language	Loss of email and VPN access
Suspicious URL	company-support-portal.com
Authentication Failure	SPF SoftFail and DMARC Fail
Social Engineering	Uses urgency to force compliance

Risk Classification

Classification: PHISHING

Risk Level: HIGH

Potential Impact

- Corporate account compromise
 - Unauthorized system access
 - VPN misuse
 - Data breaches
 - Internal network compromise
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Sample 3 – PayPal Invoice Scam

Email Scenario

The attacker impersonates a payment service provider and sends a fake invoice notification.

The goal is to create panic by claiming a payment has already been processed.

Victims are encouraged to call a fake support number or visit a malicious dispute website.

Header Analysis

Field	Value
Sender Domain	paypal-invoice-center.com
SPF	FAIL
DKIM	NONE
DMARC	FAIL
Mail Server	outbound.paypal-invoice-center.com

Phishing Indicators

Indicator	Observation
Fake PayPal Domain	Not an official PayPal domain
Financial Pressure	Unauthorized charge claim
Refund Scam	Refund available only within 24 hours
Panic Tactic	Creates urgency
Suspicious Link	Fake dispute portal
Fake Support Number	Encourages phone scam interaction

Indicator	Observation
Authentication Failure	SPF and DMARC failures

Risk Classification

Classification: PHISHING

Risk Level: HIGH

Potential Impact

- Financial fraud
- Credential theft
- Social engineering attacks
- Scam phone calls
- Personal data exposure

Comparative Analysis

Email Sample	Attack Type	Main Objective	Risk Level
Bank Account Alert	Credential Theft	Banking Credentials	High
IT Helpdesk Reset	Corporate Phishing	Employee Credentials	High
PayPal Invoice Scam	Refund Scam	Financial Information	High

Common Phishing Indicators Identified

The following indicators were observed across all analyzed phishing emails:

- ✓ Fake sender domains
- ✓ Suspicious URLs
- ✓ Generic greetings
- ✓ Urgency and time pressure
- ✓ Fear-based language
- ✓ Credential requests
- ✓ Authentication failures
- ✓ Spoofed organizational identities

- ✓ Requests for immediate action
 - ✓ Social engineering techniques
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Risk Assessment

Threat Category	Severity
Credential Theft	High
Identity Theft	High
Financial Fraud	High
Corporate Data Breach	High
Unauthorized Access	High
Social Engineering	High

Overall Risk Rating

CRITICAL

The analyzed emails demonstrate multiple indicators of phishing activity and present a significant risk to individuals and organizations. Successful exploitation could result in credential theft, financial losses, unauthorized access, and sensitive data exposure.

How These Attacks Work

Phishing attacks rely on social engineering rather than technical vulnerabilities.

Attackers impersonate trusted organizations and create messages that encourage victims to act quickly without verifying authenticity.

Typical attack flow:

1. Victim receives a phishing email.
 2. Email creates urgency or fear.
 3. Victim clicks a malicious link.
 4. Victim enters credentials or sensitive information.
 5. Attackers collect the information.
 6. Accounts are compromised.
 7. Data theft or financial fraud occurs.
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Prevention Guidelines

Do's

- Verify sender email addresses carefully.
- Hover over links before clicking.
- Confirm requests through official channels.
- Enable Multi-Factor Authentication (MFA).
- Report suspicious emails to the security team.
- Keep systems and browsers updated.
- Use strong and unique passwords.

Don'ts

- Do not click unknown links.
 - Do not open unexpected attachments.
 - Do not share passwords.
 - Do not share OTP codes.
 - Do not trust urgent requests without verification.
 - Do not provide personal information through email links.
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Employee Awareness Checklist

Before interacting with any email:

- ☐ Verify the sender domain.
 - ☐ Check for spelling or grammar mistakes.
 - ☐ Inspect hyperlinks before clicking.
 - ☐ Look for urgency or threatening language.
 - ☐ Confirm requests through official communication channels.
 - ☐ Report suspicious messages immediately.
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Conclusion

The three analyzed email samples exhibited clear phishing characteristics, including spoofed sender domains, suspicious links, urgency-based messaging, authentication failures, and attempts to collect sensitive information.

All three samples were classified as HIGH-RISK PHISHING emails due to their strong use of social engineering tactics and potential impact on victims.

Organizations should continuously educate employees, implement email security controls, enforce Multi-Factor Authentication, and encourage reporting of suspicious emails to reduce phishing-related risks.

References

- Google Message Header Analyzer
- MXToolbox Email Header Analyzer
- Future Interns Cyber Security Task 2 Guidelines
- Public Phishing Email Awareness Resources

END OF REPORT